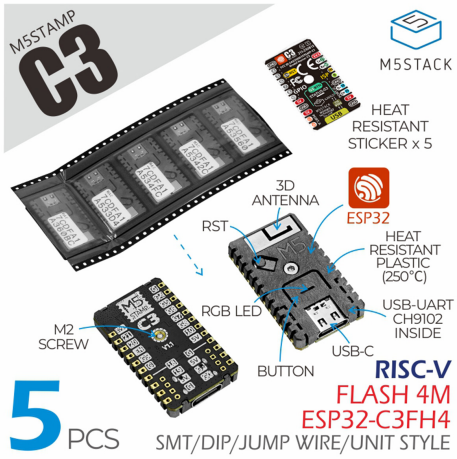
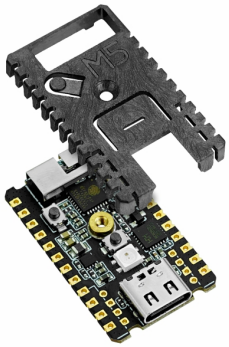
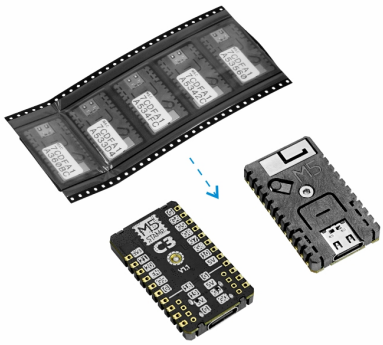
# Stamp-C3

SKU:C056-B



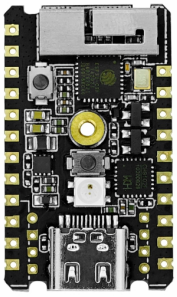




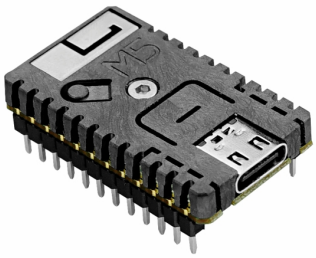














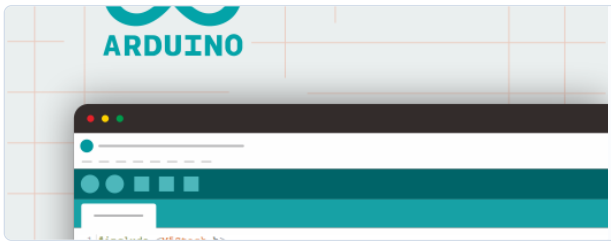





## Description

**Stamp-C3** is a cost-effective, minimal RISC-V core board powered by the **ESP32-C3** SoC from Espressif. It integrates a 32-bit single-core **RISC-V** processor running at up to 160 MHz, 400 KB SRAM, and 4 MB FLASH, together with 2.4 GHz Wi-Fi connectivity. The board supports secure boot, Flash encryption, digital signature and other security features. Despite its tiny size, it delivers robust performance to meet diverse IoT application requirements.

## Tutorial




### Arduino IDE

This tutorial will show you how to program and control the Stamp-C3 device with the Arduino IDE.

## Features

- Supports multiple mounting methods (SMT, DIP, flying leads)
- Equipped with a high-temperature plastic shield, compatible with SMT reflow (230 °C)
- On-board 5 V → 3.3 V DC/DC, 13 × GPIOs
- Type-C port
- 1 × programmable RGB LED, 1 × reset button, 1 × user button
- High-performance 3D antenna for stable, reliable wireless communication
- Smallest ESP32 system board
- Development Platform
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

## Includes

- 5 × Stamp-C3
- 5 × High-temperature Labels

## Applications

- DIY prototyping

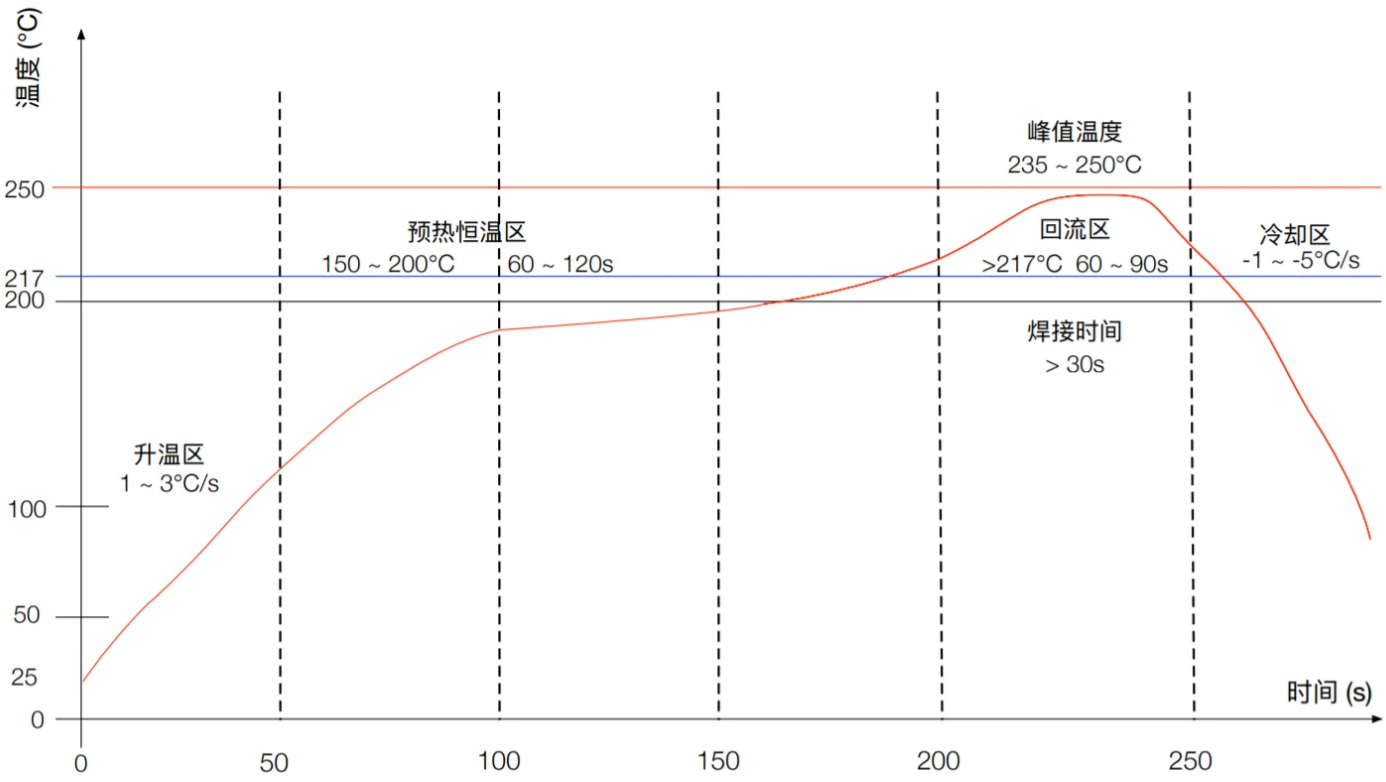
- Industrial automation
- Smart home

# Specifications

| Specification | Parameter |
| --- | --- |
| SoC | ESP32-C3@32-bit RISC-V Single-Core Processor, Main Frequency up to 160MHz |
| Flash | 4MB |
| SRAM | 400KB |
| ROM | 384KB |
| RTC SRAM | 8KB |
| Wi-Fi | 2.4 GHz Wi-Fi (supports 20/40MHz bandwidth, 802.11 b/g/n, up to 150Mbps) |
| Input Voltage | 5V@500mA |
| Human-Machine Interaction | Programmable Physical Button x 1, Reset/Debug Button x 1, Programmable RGB LED (SK6812) x 1 |
| USB Interface | USB Type-C x 1 |
| Antenna Type | 2.4G 3D Antenna |
| Module Peripheral Interface Resources | ADC, GPIO, SPI, UART, I2C, I2S, PWM, RMT, DMA, USB Serial, TWAI |
| IO Interfaces x13 | G21, G20, G9, G18, G19, G1, G0, G10, G8, G7, G6, G5, G4 |
| IO Interface Pitch | 2.54mm |
| Mounting Screw Specification | M2 x 4 Countersunk Hex Socket Machine Screw |
| Product Size | 34.0 x 20.0 x 4.6mm |
| Single Product Weight | 4.1g |
| Package Size | 138.0 x 93.0 x 5.6mm |
| Gross Weight | 23.3g |

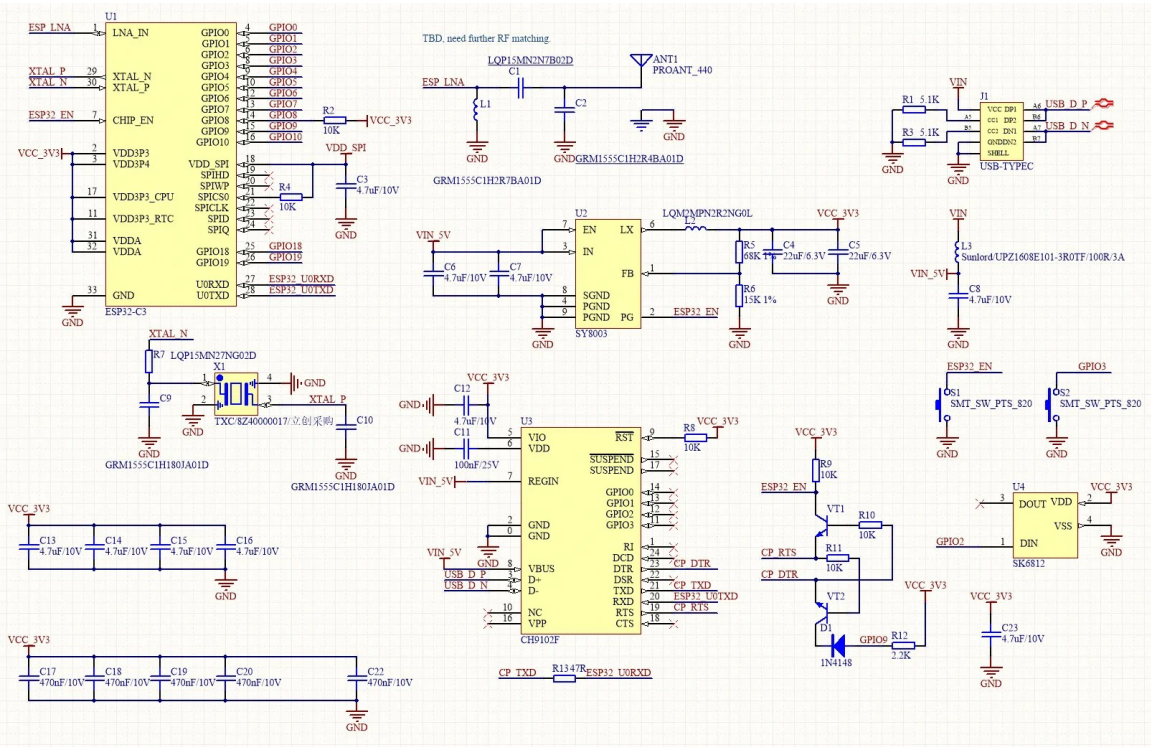
# Learn

## Reflow Profile Supported by the Enclosure

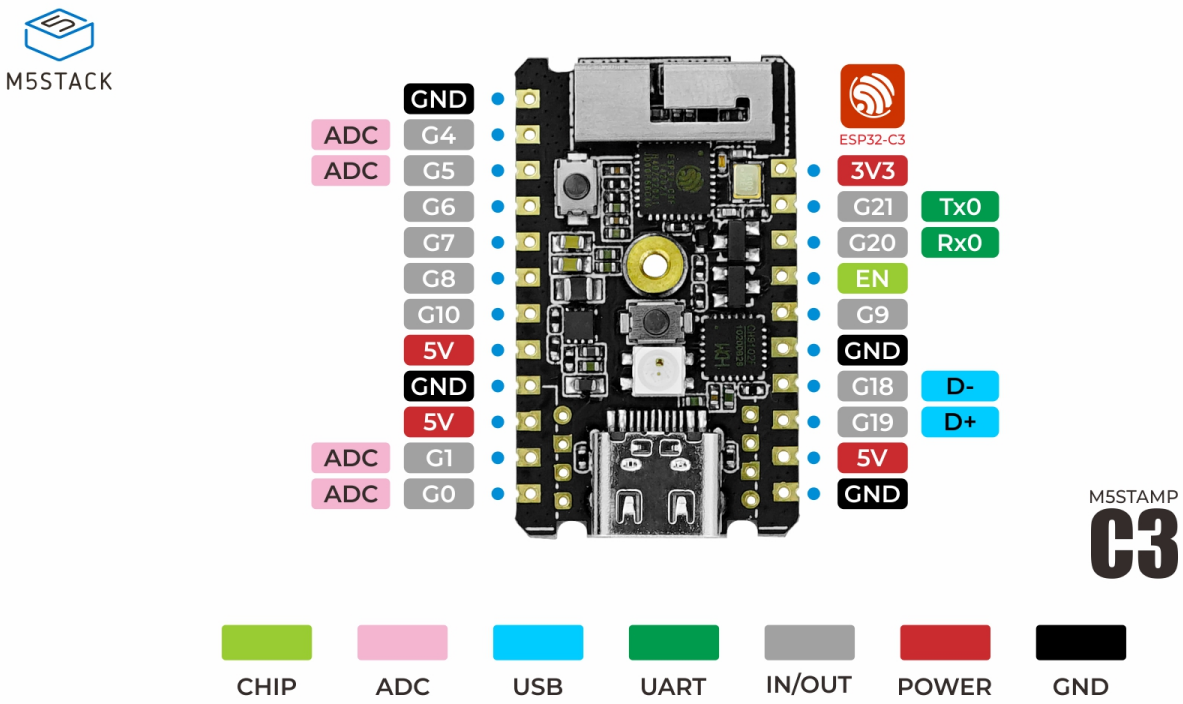



升温区 — 温度：25 ~ 150°C 时间：60 ~ 90s 升温斜率：1 ~ 3°C/s
预热恒温区 — 温度：150 ~ 200°C 时间：60 ~ 120s
回流焊接区 — 温度：>217°C 时间：60 ~ 90s；峰值温度：235 ~ 250°C 时间：30 ~ 70s
冷却区 — 温度：峰值温度 ~ 180°C 降温斜率 -1 ~ -5°C/s
焊料 — 锡银铜合金无铅焊料 (SAC305)

## Schematics



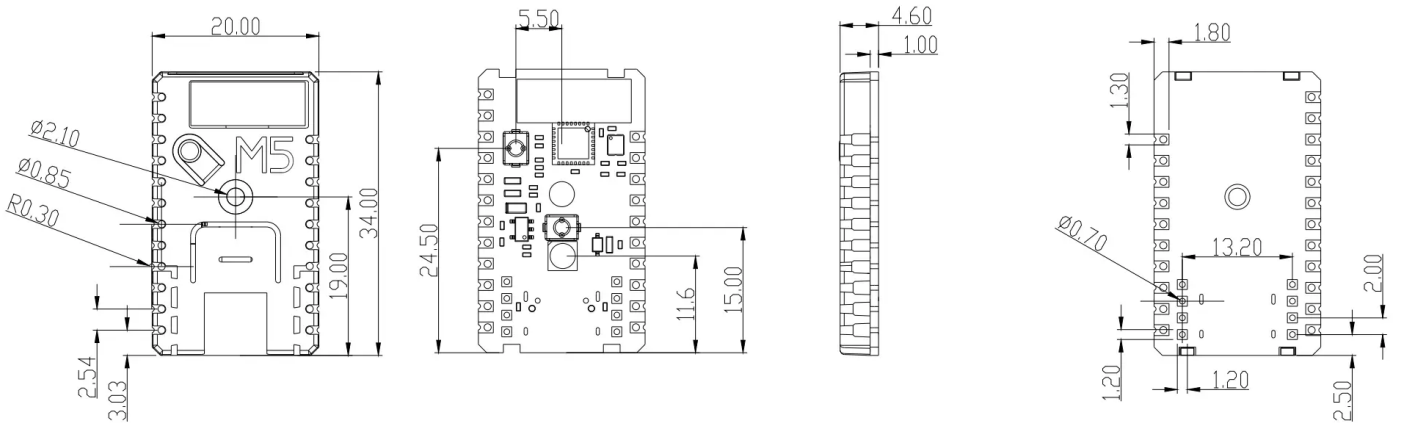
# PinMap




## SK6812 & Button

| ESP32-C3 | G2 | G3 |
|---|---|---|
| SK6812 | DI | / |
| Button | / | SW |

# Model Size




UNIT: mm

# Structure

- Stamp-C3 Structure Files

# PCB

- Stamp-C3 KiCad Library
- Stamp-C3 PcbDoc

# Datasheets

- ESP32-C3

# Softwares

## Arduino

- Stamp-C3 Arduino Quick Start
- Stamp-C3 Test Program

## ESP-IDF

- Stamp-C3 RGB LED Control Example

## USB Driver

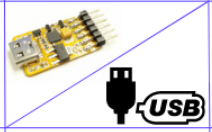
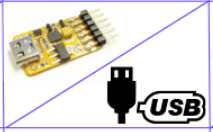
Click the link below to download the driver for your operating system. Choose the installer that matches your OS bit version. (If you are unsure which USB chip your device uses, you can install both drivers. **CH9102_VCP_SER_MacOS v1.7** may report an error during installation, but the driver is actually installed—just ignore the message.) If you encounter issues such as download timeout or "Failed to write to target RAM," try reinstalling the driver.

| Driver Name | Supported Chip | Download |
| --- | --- | --- |
| CH9102_VCP_SER_Windows | CH9102 | Download |
| CH9102_VCP_SER_MacOS v1.7 | CH9102 | Download |

# Product Comparison

M5STAMP

Comparison chart

| COMPARISON | STAMP PICO | STAMP C3U | STAMP C3 | STAMP S3 |
|---|---|---|---|---|
| PICTURE |  |  |  |  |
| Number of IO ports | 12 | 14 | 13 | 23 |
| FLASH | 4M | 4M | 4M | 8M |
| Serial IC | / | USB CDC | CH9102F | USB CDC |
| Download method |  |  |  |  |
| Pin spacing | 2.54mm | 2.54mm | 2.54mm | 2.54mm/ 1.27mm |
| LCD interface | ✗ | ✗ | ✗ | ✓ |
| CPU frequency | 240MHz | 160 MHz | 160 MHz | 240MHz |
| Size | 24 × 18 × 4.6mm | 34 × 20 × 4.6mm | 34 × 20 × 4.6mm | 26 × 18 × 4.6mm |

To compare information on the Stamp series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.